Bounded-Authority Agent Specialization for Adaptive Technical Interview Systems: Coordination, Safeguards, and Empirical Trade-offs

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Abstract. Adaptive technical-interview systems increasingly combine automated assessment, reinforcement learning, and language-model generation, but a collection of modules is not automatically a useful multi-agent system. PrepAIred is studied here as a centrally coordinated architecture with bounded Strategy and Coaching agents: Strategy maintains a persistent weakness ledger and proposes difficulty or probe actions, while Coaching maintains coaching history and proposes feedback changes. A deterministic coordinator validates typed proposals, enforces authority boundaries, arbitrates against a frozen PPO policy, and commits state. Coordination was evaluated by deterministic replay of eight recorded answer sets under twenty question-order permutations, yielding 160 sessions per condition. Relative to the frozen baseline, the full configuration changed the difficulty path in all 160 sessions and changed mean level changes by -2.67 (cluster-bootstrap 95% CI $[-4.50, -0.24]$) and reversals by -1.88 ($[-3.12, -0.27]$). However, split and unified controllers made identical decisions over 1,280 matched turns, so decomposition itself was not shown to improve decisions. Fault injection found 38/40 adversarial-valid sessions with guardrail bypass without arbitration versus 0/40 with arbitration, while a grounding probe accepted unsupported paraphrased credit in 57/57 unambiguous cases. A short evaluator study found $\rho = 0.4832$ for the CrossEncoder signal, $\rho = 0.3812$ for the deployed composite, and a word-count baseline of $\rho = 0.4897$; the length-controlled partial correlation was 0.2434 (95% CI $[-0.008, 0.507]$). The evidence therefore supports bounded authority and property-specific safeguards as architectural mechanisms, not a general claim of multi-agent superiority or pedagogical benefit.

Keywords: multi-agent systems · bounded authority · agent coordination · technical interview assessment · reinforcement learning · fault injection

1 Introduction

Technical interviews require candidates to demonstrate domain knowledge, structured reasoning, problem solving, and clear communication in a short interactive

session. Automated preparation systems address different parts of this problem: static or rubric-based assessment emphasizes reproducibility, conversational LLM interviewers emphasize flexible dialogue, adaptive tutoring emphasizes sequencing, and multi-agent educational systems emphasize role specialization. The systems question studied here is narrower: when these capabilities are combined, which component is allowed to make which decision, how are specialized recommendations coordinated, and how can the resulting behavior be tested?

Automated short-answer assessment has moved from lexical matching toward semantic representations and entailment-oriented scoring [1,2]. Speech-enabled interview systems additionally use acoustic and multimodal signals for interview-related prediction [4,5,6]. Adaptive tutoring and testing use knowledge tracing, item-response methods, and reinforcement learning to select instructional content [9,10]. Large language models add flexible generation but also create risks of unsupported generation and model-dependent judging [7,17]. Multi-agent frameworks such as AutoGen and MetaGPT demonstrate conversational or role-based decomposition [8,11], while recent work reports failure modes involving specification drift, coordination, and verification [3]. These lines of work solve complementary parts of the problem, but they do not establish whether agent decomposition itself changes decisions inside an already-deployed adaptive technical-interview controller.

PrepAIred addresses this systems question by distinguishing computational services from explicit weak agents. A deterministic technical evaluator owns numerical technical scoring; an independent communication-confidence component produces a separate state signal; a frozen PPO policy proposes difficulty; a question selector instantiates valid questions; and a local Qwen2.5-1.5B-Instruct model provides grounded natural-language follow-up and feedback. The specialization configuration then adds two stateful agents: a Strategy Agent that maintains a weakness ledger and proposes difficulty/probe actions, and a Coaching Agent that maintains a coaching-history ledger and proposes presentation changes. Neither agent writes committed state. The coordinator validates proposals, checks authority, version, and evidence conditions, arbitrates them against deterministic rules and the frozen PPO action, and remains the sole system of record.

The architecture is intentionally centrally coordinated rather than decentralized multi-agent reinforcement learning. The objective is not to maximize agent autonomy, but to make role specialization explicit while preserving least-privilege authority, auditable state ownership, and reproducible behavior. The contribution is therefore evaluated at three levels: coordination effects, specialization limits and operational trade-offs, and property-specific safeguards. A short evaluator-validity study is included because evaluator outputs are inputs to Strategy, but evaluator validation is not the paper's principal contribution.

The research questions are:

1. **RQ1 – Coordination:** Does the bounded agent layer change interview difficulty trajectories relative to the frozen single-controller baseline?

2. **RQ2 – Specialization:** Does a split Strategy/Coaching architecture produce different decisions from a functionally equivalent unified controller under matched observations and rules?
3. **RQ3 – Safeguards:** Which implemented validation, grounding, arbitration, threshold, and fault-containment mechanisms are necessary for the measured safety properties?
4. **RQ4 – Evaluator validity:** How closely does the frozen technical evaluator agree with the available human-reference benchmark, and what is known about length dependence?

The paper makes three primary contributions. First, it specifies a bounded-authority architecture with private ledgers, typed proposals, explicit authority contracts, and deterministic arbitration. Second, it reports replay and fault-injection evidence, including negative findings, that separates architectural decomposition from measured decision benefit. Third, it reports the available evaluator-validity evidence and its length confound. Bloom-style question metadata are retained as descriptive structure only; no experiment here shows that Bloom labeling improves adaptive difficulty or learning.

2 Related Work

2.1 Automated short-answer and interview assessment

Sentence-BERT provides efficient semantic sentence representations [1], while automated short-answer grading has combined semantic similarity and entailment [2]. Interview-oriented work has used acoustic and multimodal features to model delivery or interview outcomes [4,5,6]. PrepAIred combines semantic similarity, rubric concept coverage, and a CrossEncoder-derived signal for technical scoring while keeping communication confidence as a separate state variable.

2.2 LLM judging and grounded generation

LLMs can provide flexible feedback, but hallucination and model-dependent evaluation behavior remain known concerns [7,13,17]. Recent evidence-grounded evaluation work motivates separating evidence computation from free-form generation. PrepAIred therefore gives the local language model a downstream generation role rather than numerical scoring or difficulty authority.

2.3 Adaptive tutoring and runtime enforcement

Knowledge tracing and cognitive-tutor approaches model learner state for sequencing [9,10]. PPO provides a practical policy-gradient method for continuous observations and discrete actions [12,14]. Safe reinforcement-learning work studies external shielding and runtime constraints [1,19]. PrepAIred similarly places deterministic guardrails and arbitration around a frozen PPO action; the Strategy and Coaching agents are not trained and the system is not presented as MARL.

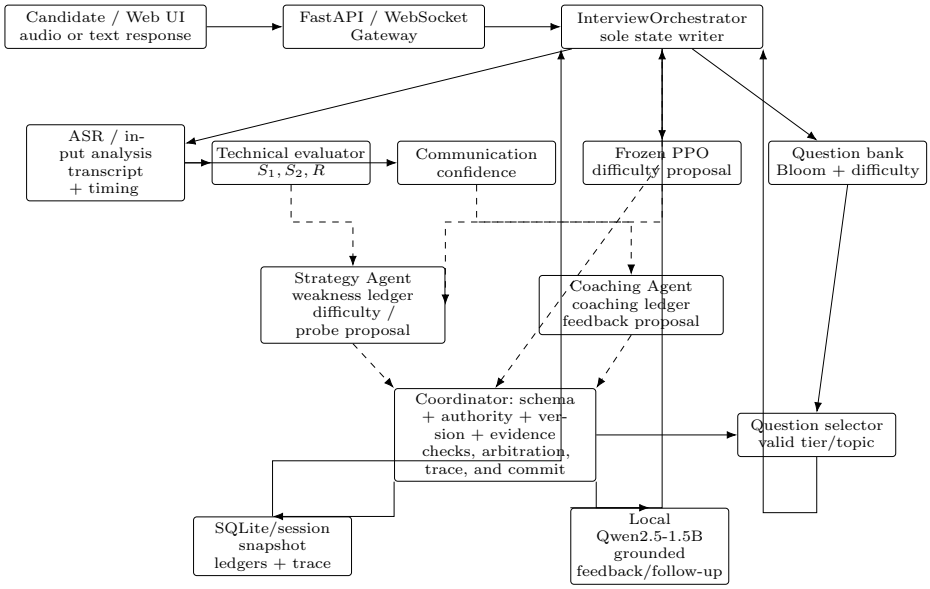


Fig. 1. PrepAIred runtime architecture. The evaluator, communication model, frozen PPO policy, question bank, selector, and local LLM are specialized system components; Strategy and Coaching are the explicitly evaluated weak agents. Dashed arrows denote typed proposals. The coordinator is the sole state writer and applies validation, authority checks, arbitration, and trace recording.

2.4 Multi-agent coordination and specialization

Classical agent research characterizes agents through autonomy, reactivity, proactivity, and social behavior [16]. Modern LLM agent frameworks include conversational coordination and role-based workflows, such as AutoGen and MetaGPT [8,11]. Comparative work on agent configurations reports task-dependent rather than universal gains [9,10]. The present study asks a more specific systems question: whether specialization changes decision and trajectory behavior when two controllers receive matched information and implement matched rules.

3 System Architecture and Bounded Authority

The system contains an interview interface, FastAPI/WebSocket gateway, orchestration hub, ASR and response-analysis services, a hybrid technical evaluator, an independent communication-confidence component, a frozen PPO policy, a structured question bank and selector, and a local Qwen2.5-1.5B-Instruct generator served from GGUF/llama.cpp. The agent layer adds Strategy and Coaching roles around this existing hub.

Table 1. Bounded-authority matrix in the specialization configuration. Neither agent can write committed state; the hub commits alone.

Role	Private state	May propose	May not do
Strategy	Weakness ledger, miss counts, probe history, score trend	Difficulty action; probe request; strategy mode	Write score; set difficulty directly; edit history; terminate session; override hard constraints/guardrails; call another agent
Coaching	Coaching history, hints/examples, repeated misconceptions	Feedback depth; hint/example flags; emphasis concepts; remediation tag; tone	Alter score/evaluator evidence; request difficulty changes; request follow-ups; terminate session
Coordinator	Committed session state and versions	Arbitrated action/plan; trace records	Delegate sole state ownership
Qwen generator	Grounded prompt context and evaluator evidence	Follow-up wording; qualitative feedback	Determine score, grade, or difficulty

3.1 Strategy Agent

The Strategy Agent maintains a private weakness ledger containing per-concept misses, probe history, and a score trend. Its first-match rules are: recover when current and recent performance are low; remediate a repeated unresolved concept by proposing 'Same' plus a probe; stretch when sustained high performance and no open weakness are observed; otherwise remain steady. Non-steady confidence is a heuristic function of cited evidence count and is not a calibrated probability. Steady proposals are assigned confidence 0.45 and therefore do not supersede PPO at $\tau = 0.6$.

3.2 Coaching Agent

The Coaching Agent maintains a private coaching history containing prior hints/examples and repeated misconceptions. It can propose feedback depth, up to two concepts to emphasize, hint/example flags, tone, and a remediation tag. Strategy's peer message can condition presentation (for example, detailed during recover or remediation emphasis during a targeted probe), but Coaching cannot change the evaluator score, request a difficulty change, request additional follow-ups, or terminate the session.

3.3 Authority matrix

3.4 Typed communication and state ownership

Every observation, proposal, and decision is typed. Unknown fields, NaNs, out-of-range values, stale versions, and unauthorized state fields are rejected. The coordinator records session, turn, sender, recipient, state version, provenance, and an accept/reject/modify reason. An agent timeout or unhandled exception causes its proposal to be dropped; the default policy continues.

3.5 Question bank and Bloom metadata

Each question record carries topic, numeric difficulty, Bloom-style cognitive level, question type, and time-limit metadata. The operational Bloom bands start at Understand and proceed through Apply, Analyze, Evaluate, and Create. These labels are descriptive question metadata, not a tested educational intervention. The system also refers to coarse Easy/Medium/Hard pools while using a five-level numeric difficulty representation; the supplied evidence does not establish a validated mapping between the two representations, so no efficacy claim is made for that mapping.

4 Orchestration, Arbitration, and Validation

The implemented precedence is deterministic. Hard constraints establish legal state/action bounds. The guardrail layer is applied last and wins when a rule fires. A validated Strategy proposal can supersede PPO when it satisfies authority, evidence, version, and confidence requirements. Otherwise the frozen PPO action, or the labeled non-RL fallback when PPO is unavailable, remains the candidate action.

The principal arbitration threshold is $\tau = 0.6$. Difficulty levels remain in $[1, 5]$ and the RL-phase transition is bounded to one level. Probe precedence is bounded by at most two consecutive follow-ups; a validated Strategy probe may add or retarget a follow-up when its cited concept is grounded and its confidence reaches τ; otherwise the original threshold-based hub rule remains responsible.

4.1 Guardrail implementations

The deployed hub uses an inline copy of the rule layer that differs from the training/simulation module at documented boundary states. On a synthetic grid of 39,930 states, the two implementations disagree at 4,728 points (11.8

4.2 One-turn protocol

5 Technical Evaluator and Communication Model

5.1 Technical evaluator

Let S_1 denote sentence-level semantic similarity, S_2 rubric concept coverage, and R the normalized CrossEncoder signal. The base score is

$$S_{\text{base}} = 0.15S_1 + 0.35S_{2,\text{eff}} + 0.50R. \tag{1}$$

Concept coverage is damped when reasoning evidence is weak:

$$S_{2,\text{eff}} = \begin{cases} S_2, & R > 0.30, \\ 0.60S_2, & R \leq 0.30. \end{cases} \tag{2}$$

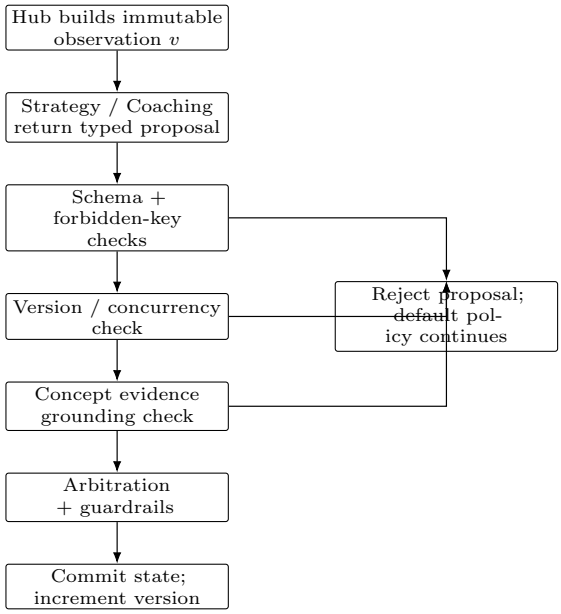


Fig. 2. Proposal-validation pipeline. Each proposal is checked before arbitration; rejection does not mutate committed state.

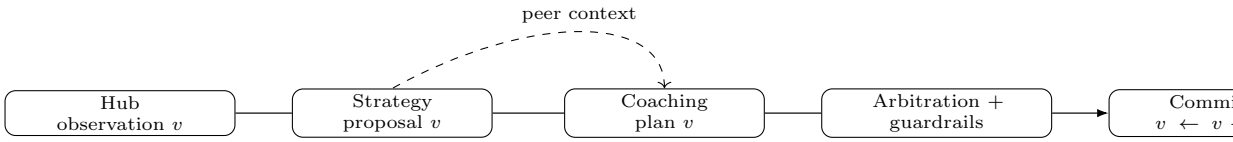


Fig. 3. A verbal turn is organized as observation, typed proposals, validation, arbitration, and commit. The coordinator remains the only state writer.

The CrossEncoder output is normalized using the reported regression mapping

$$R = \mathrm{clamp}\left(\frac{r_{\mathrm{CE}} - 0.20}{0.70}, 0, 1\right), \tag{3}$$

where r_{CE} is the saved regression output. Bonus matches add 0.03 when their similarity exceeds 0.50; explicit mistake-vector matches incur 0.07 each, capped at 0.30. The reported weak-evidence terms are

$$P_{\mathrm{weak}} = 0.20\,\mathbf{1}[R \le 0.25 \wedge S_2 \le 0.25] + 0.05\,\mathbf{1}[R < 0.30 \wedge S_2 < 0.50], \tag{4}$$

and the documented hedge term is

$$P_{\mathrm{hedge}} = 0.03\,\mathbf{1}[S_2 < 0.50 \wedge R < 0.40]. \tag{5}$$

With P_{mistake} denoting the capped misconception penalty, the final reported formulation is

$$S_{\text{final}} = \text{clamp}\left(S_{\text{base}} + S_{\text{bonus}} - P_{\text{mistake}} - P_{\text{weak}} - P_{\text{hedge}}, 0, 1\right). \tag{6}$$

Missing mandatory concepts impose a 0.60 score ceiling. This evaluator is authoritative for technical scoring; the LLM does not overwrite the score.

5.2 Communication confidence

Communication is a separate state variable. The documented acoustic terms are

$$C_j = \text{clamp}(1 - j/0.03), \tag{7}$$
$$C_s = \text{clamp}(1 - s/0.12), \tag{8}$$
$$C_h = \text{clamp}(h/28), \tag{9}$$
$$V_Q = 0.32C_j + 0.30C_s + 0.28C_h + 0.10C_{pitch}. \tag{10}$$

Fluency, linguistic confidence, speaking rate, and rhythm are combined into C_{final}. The configured speaking-rate band is 2.1–4.3 words/s. The score is a computational proxy, not a validated psychological measure, and the supplied evidence does not establish fairness across accents, environments, or disabilities.

6 Adaptive Difficulty and Question Selection

PrepAIred represents adaptive difficulty as a six-dimensional normalized state consisting of current performance, recent performance, communication confidence, hesitation, normalized session progress, and current difficulty. The action space is $\{\text{Easier}, \text{Same}, \text{Harder}\}$. The learned PPO checkpoint is frozen during coordination evaluation. A deterministic initial-question phase precedes policy-driven adaptation, and the selector enforces topic/difficulty constraints and question non-repetition.

The question bank also stores Bloom-style cognitive levelization as descriptive metadata. Because no experiment compares Bloom-aware selection against a non-Bloom control, the paper does not claim that Bloom improves learning, adaptation, or difficulty calibration.

7 Methodology

7.1 Evidence classes

Four evidence classes are distinguished: **REPLAY**, deterministic replay through the real orchestrator with recorded evaluator outputs; **FAULT-INJECTED**, controlled removal or corruption of safeguards; **SOCKET**, boundary-level interface checks; and **SYNTHETIC**, controlled state-grid or parameter probes.

7.2 Replay structure

The coordination study uses eight recorded answer sets and twenty question-order permutations, yielding $8 \times 20 = 160$ sessions per condition. These sessions reuse the same eight structural clusters and are therefore dependent. Cluster bootstrap resamples the eight clusters rather than treating the 160 permutations as independent sessions. Bootstrap intervals use 10,000 resamples; no p-values are reported. The replay order seeds are 42–61 in the documented runs.

No live generative LLM is invoked during replay. The LLM gateway is stubbed as unavailable, and deterministic renderers provide text outputs so that trajectory effects are attributable to the tested coordination mechanisms rather than model sampling.

7.3 Experimental conditions

The main replay conditions are B0 (frozen PPO baseline), B7 (PPO-off single-controller fallback), B0c (canonical guardrails), M1 (full bounded-agent configuration), M2 (Strategy ablated), M3 (Coaching ablated), M4 (arbitration removed), M5 (grounding verification removed), M6 (Strategy-to-Coaching messaging removed), M7 (PPO-off multi-agent), and M8 (authority enforcement ablated). The split-versus-unified comparison uses the two-agent architecture and a monolithic joint-ledger controller receiving matched observations, thresholds, and arbitration rules.

7.4 Fault-injection design

Faults cover crash/exception, timeout, malformed output, forbidden keys, stale versions, NaNs/out-of-range values, adversarial-valid proposals, and ungrounded concepts. The fault-injection configuration uses a fault rate of 0.3 in the corresponding 40-session experiments. Binary containment outcomes use exact Clopper–Pearson intervals.

8 Results

8.1 XA: coordination dynamics

All evaluated conditions completed 160 sessions. Committed scores remained equal to recorded evaluator outputs and difficulty stayed within [1,5] with one-step RL-phase transitions in every session. The exact Clopper–Pearson lower bound for a 160/160 property is 0.977.

M1 altered the difficulty trajectory in all 160 sessions. Relative to B0, mean level changes differed by -2.669 with cluster-bootstrap 95% CI $[-4.50, -0.24]$, and reversals differed by -1.875 with CI $[-3.12, -0.27]$. PPO agreement changed by -0.226 with CI $[-0.284, -0.170]$. The direction is consistent with reduced trajectory volatility in this replay setting, not with verified improvement in learning or candidate outcomes.

Table 2. Replay ablation summary across 160 sessions per condition. Mean trajectory and latency quantities are per session unless noted otherwise.

Cond.	Compl.	Trans.	Chg.	Rev.	PPO agr.	Strat.	Bypass	p50 ms
B0	160	160	15.119	11.744	0.626	0.000	0	0.546
B7	160	160	2.675	1.256	–	0.000	0	0.463
B0c	160	160	15.119	11.744	0.626	0.000	0	0.553
M1	160	160	12.450	9.869	0.400	6.669	0	1.589
M2	160	160	15.119	11.744	0.626	0.000	0	1.218
M3	160	160	12.450	9.869	0.400	6.669	0	1.118
M4	160	160	2.225	1.000	0.014	19.619	3	1.573
M5	160	160	12.450	9.869	0.400	6.669	0	1.581
M6	160	160	12.450	9.869	0.400	6.669	0	1.592
M7	160	160	2.706	1.312	–	9.037	0	1.502
M8	160	160	12.450	9.869	0.400	6.669	0	1.575

B7 disables PPO and the agent layer and therefore produces only 2.675 level changes per session. This comparison shows why lower volatility is not itself a quality criterion: a much smoother trajectory can result from removing the learned adaptation mechanism.

The Strategy Agent generated 3,814 proposals across replay and emitted zero stretch proposals. The +0.59 additional probe turns per session were entirely driven by replay cluster $k = 1$; the other seven clusters produced no extra probes. The M2 condition reproduces B0, while M3, M5, M6, and M8 reproduce M1's trajectory under the matched replay configuration.

8.2 XB: fault injection and safeguard properties

Every fault-injection cell preserved score authority and valid transitions in 160/160 sessions. Because this property is at the ceiling, it does not discriminate among safeguards.

Schema validation alone contained forbidden-key output in the corresponding injected test. For the adversarial-valid fault, removing arbitration yielded guardrail bypass in 38/40 sessions, versus 0/40 with arbitration. The corresponding exact 95% Clopper–Pearson intervals are $[0.831, 0.994]$ and $[0.000, 0.088]$, respectively. These counts refer to sessions.

The per-agent breaker opened in 29/40 sessions (72.5%), below the predicted 80% threshold. The exact 95% Clopper–Pearson interval is approximately $[0.561, 0.854]$. The injected fault rate was 0.3, i.e., 12/40 sessions; its exact 95% interval is approximately $[0.166, 0.465]$. Pooled p95 latency was about 154 ms, close to the configured 150 ms timeout.

For ungrounded emphasis, removing the grounding check produced the targeted failure in 40/40 sessions, while retaining the check produced 0/40. A well-formed but semantically wrong proposal was not detected by structural validation; thus syntax, authority, version, and lexical evidence checks do not constitute semantic correctness verification.

Table 3. Safeguard-specific fault results. Global containment is reported separately because it is at the ceiling in every condition.

Mechanism	Observed property
Schema validation	Forbidden-key fault contained; 160/160 global containment in every cell
Authority check	Unauthorized protected fields rejected; scores remain authoritative
Version check	Stale/duplicate proposals are dropped before arbitration
Grounding check	40/40 targeted ungrounded failures without check versus 0/40 with check
Confidence threshold	$\tau = 0$ alone reproduces M4 trajectory collapse
Arbitration	Adversarial-valid bypass 38/40 without versus 0/40 with arbitration
Guardrails	Legal bounds preserved; global containment at ceiling
Per-agent breaker	Opens in 29/40 injected sessions; below predicted 80% threshold

8.3 XC: threshold decomposition

The confidence threshold was swept over $\tau \in \{0.0, 0.5, 0.6, 0.7, 0.8, 0.9\}$ while other evidence checks and guardrails remained active. At $\tau = 0.0$, mean level changes collapsed to 2.225 and PPO agreement to 0.014, reproducing the qualitative M4 trajectory despite retaining the other safeguards. At $\tau = 0.5, 0.6, 0.7$, level changes were 12.450 and PPO agreement 0.400; at $\tau = 0.8$, changes remained 12.450 and agreement was 0.404. At $\tau = 0.9$, Strategy proposals were silenced and the trajectory returned to the B0 values 15.119 changes and 0.626 PPO agreement.

The realised Strategy confidence values were only 0.45, 0.70, and 0.80. Therefore the apparent plateau is not robustness to a continuously varying threshold; it is a consequence of discrete observed confidence levels.

8.4 XD: grounding probe

The lexical grounding check accepted unsupported paraphrased credit in 57/57 unambiguous cases. Seventeen earlier apparent rejections were attributed to artifacts of the check. The result therefore constrains the interpretation of grounding: the current mechanism can enforce a specific lexical/evidence contract, but its semantic verification ability is not established.

8.5 Contradicted predictions

8.6 Specialization: split versus unified

A turn is one replayed candidate-response/evaluator-state update. The matched split-versus-unified study uses 64 recorded answer instances across 20 question-order permutations, yielding 1,280 turn-level observations grouped into 160 sessions and eight replay clusters. The unified controller is a joint-ledger reimplementation of the same rule set and receives identical observations, question orders, thresholds, and arbitration rules.

Table 4. Contradicted predictions retained in the merged analysis.

ID	Prediction	Outcome
A9	Disabling PPO would change at least 50% of session paths	4/160 (2.5%); contradicted
A10	Canonical guardrails would change at least one path vs. legacy	0/160; contradicted
B6	Per-agent breaker would open in at least 80% of injected sessions	29/40 (72.5%); contradicted

The split and unified architectures produced identical normalized decision outputs over all 1,280 turns: $P(\text{Decision}_{unified} = \text{Decision}_{split}) = 1.0$. The normalized decision tuple includes Strategy difficulty action, mode, confidence, probe concept, and Coaching depth, hint/example flags, tone, remediation tag, and emphasis concepts. Because the two implementations execute the same deterministic rules under matched inputs, this is a consistency check rather than evidence of superiority or necessity.

Removing Coaching also leaves difficulty trajectories unchanged because Coaching has no information path back to the difficulty arbitration logic. Peer messaging likewise changes only coaching presentation in the tested architecture. Split versus unified component dispatch is 0.0440 ms versus 0.0401 ms, and the two-model serialization micro-benchmark is 0.0092 ms versus 0.0038 ms. The broader multi-agent layer adds +1.043 ms median per-turn latency relative to B0 (1.589 ms versus 0.546 ms).

9 Evaluator Validity

The evaluator study is short because evaluator validity is not the paper's main contribution. The benchmark contains 64 constructed technical answers across eight questions and an available human-reference score. The CrossEncoder-only signal gives Spearman $\rho = 0.4832$; the deployed composite gives $\rho = 0.3812$; a word-count baseline gives $\rho = 0.4897$. The CrossEncoder is associated with answer length, and the length-controlled partial correlation is 0.2434 with 95% CI $[-0.008, 0.507]$.

The paired contrast between the CrossEncoder-only and composite correlations includes zero. Consequently, the benchmark does not establish composite superiority or a length-independent reasoning contribution from the CrossEncoder signal. Rater provenance is incomplete and the benchmark is not treated as independent expert-panel validation.

10 Discussion

10.1 What the evidence shows

The architecture provides explicit authority separation, private state, typed communication, and deterministic arbitration. The empirical evidence shows that the full agent layer changes replayed difficulty trajectories relative to B0, and the ablations associate this effect with the Strategy proposal/arbitration path. However, the matched split-versus-unified comparison shows no decision divergence, and Coaching or peer-message removal does not change the difficulty trajectory.

Thus, the strongest claim is not that multi-agent decomposition improves interview adaptation. Instead, the evidence supports a narrower systems claim: a centrally coordinated, bounded-authority architecture can make role specialization and state ownership explicit while allowing property-specific safeguards to be isolated and tested.

10.2 Design lessons

1. **Architectural decomposition does not imply decision benefit.** Split and unified controllers were identical under matched deterministic rules.
2. **Threshold sweeps need informative confidence variation.** The 0.45/0.70/0.80 confidence values make the observed plateau a discrete-level effect rather than a robustness result.
3. **Ceiling containment metrics need property-specific tests.** The 160/160 global invariant is valuable but non-discriminating; adversarial-valid and grounding tests discriminate between mechanisms.
4. **Lexical grounding is not semantic grounding.** The 57/57 probe demonstrates a concrete weakness.
5. **Lower trajectory volatility is not a quality metric.** B7 is much more stable than B0 while disabling PPO and the agent layer.

11 Threat Model and Limitations

All coordination experiments use deterministic replay with eight underlying clusters reused across twenty question orders, so the effective structural sample remains eight clusters. The +0.59 proactive-probe increase occurs only in cluster $k = 1$. The M1–B0 comparison uses one frozen PPO checkpoint; the observed baseline volatility is not a claim about PPO in general. The Strategy Agent never produced a Harder proposal in replay, so that branch is unexercised.

No live generative LLM is executed during the coordination replay. Candidate responses are not dynamically regenerated after a changed question difficulty, so the study cannot establish candidate learning, pedagogical benefit, satisfaction, or placement success. The evaluator benchmark is exploratory and uses constructed answers with incomplete rater provenance. The communication-confidence signal is not human-validated and makes no fairness or psychometric claim.

Table 5. Fault model used in the fault-injection analysis.

Fault	Scope
Crash / exception	Agent terminates unexpectedly during a turn
Timeout	Agent exceeds the 150 ms defensive timeout
Malformed proposal	Schema-invalid, NaN, or out-of-range output
Forbidden key	Proposal attempts to write protected state
Stale proposal	Proposal carries an obsolete state/session version
Ungrounded concept	Proposal cites unsupported concept evidence
Adversarial-valid	Proposal is syntactically valid and authorized but semantically wrong
Compromised hub	Out of scope
Malicious agent code	Out of scope below the proposal interface

The 57/57 grounding result shows a semantic limitation of the lexical evidence check, and well-formed but semantically wrong proposals remain possible. The exact software-library versions associated with the evaluator score generation are not specified in the supplied evidence; therefore the paper does not make dependency-specific reproducibility claims at that granularity. The supplied materials also do not provide a validated mapping between the coarse Easy/Medium/Hard pools and the five numeric difficulty tiers.

No formal institutional ethics review was obtained for the retained replay corpus. The replay payloads are described as containing no names, direct identifiers, or sensitive personal information and are used only as offline system-evaluation inputs.

12 Reproducibility

The paper reports deterministic replay, cluster-aware bootstrap with 10,000 resamples, and local experiment tags. The split-versus-unified comparison uses identical observation inputs, thresholds, and question orders. The retained evidence supports release of anonymized replay inputs and experiment scripts subject to repository and privacy constraints. Exact dependency manifests should accompany any public release.

13 Conclusion

PrepAIred provides a centrally coordinated bounded-authority architecture in which Strategy and Coaching maintain distinct ledgers and produce typed proposals while the coordinator remains the sole state writer. The evidence is intentionally mixed. The full configuration changes replayed difficulty trajectories relative to the frozen PPO baseline, but a functionally equivalent unified controller produces identical decisions over 1,280 matched turns. The architecture therefore represents a coordination and authority mechanism rather than a demonstrated decision-quality advantage from agent decomposition.

The most discriminating empirical findings are property-specific. Arbitration eliminates the tested adversarial-valid guardrail bypasses; the grounding check

blocks the tested ungrounded-emphasis faults; and the core state invariants remain intact across the evaluated fault matrix. At the same time, the grounding probe accepts unsupported paraphrased credit in 57/57 unambiguous cases, and well-formed but semantically wrong proposals are not detected by structural validation. The evaluator study likewise does not establish a length-independent reasoning contribution.

Future work should evaluate stronger single-agent baselines, closed-loop candidate dynamics, stronger semantic grounding, live-LLM fault behavior, evaluator weight sensitivity, and human-participant outcomes. These are future protocols rather than current results.

Disclosure of Interests

The authors have no competing interests to declare relevant to this work.

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